

Degree-based topological indices measure molecular size: a null-baseline protocol for QSPR

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Abstract

In QSPR studies using degree-based topological indices, the reported coefficient of determination and the descriptor’s actual contribution move in *opposite* directions: as R^2 rises from 0.34 to 0.94, the improvement over simply counting heavy atoms collapses from +0.148 to +0.005. We explain this. Writing a degree-based index as $\text{TI}_f(G) = m\langle p(G), f \rangle$, with m the edge count and $p(G)$ the edge-degree-pair profile, we bound the correlation between any such index and m in terms of three quantities: the variation of the weighting function along directions in which real molecular profiles differ, the mean edge weight, and the coefficient of variation of molecular size in the sample. A corollary quantified over *all* properties follows: no degree-based index can differ from edge count in its correlation with *any* property by more than $\sqrt{2(1-\rho)}$, where ρ is its correlation with edge count. On 8,583 drug-like molecules the bound is valid for all 14 indices tested and tight to within 0.013 for ten; the corollary holds without exception across 70 index–property pairs. Empirically the 14 indices span one dimension (PC1 = 97.5% on 6,000 ZINC molecules). Against molecular weight no index improves on heavy-atom count on any dataset tested; against measured melting points the best single index reaches $R^2 = 0.157$, below a two-number baseline of atom count plus ring count (0.218). At $N \approx 15$ a permutation null places the 95th percentile at $R^2 \approx 0.35$ for linear fits, 0.50 for quadratic and 0.63 for cubic — so raising the fit order raises the bar a claim must clear rather than lowering it. We recommend a referee-facing checklist whose central element is a single diagnostic: report ΔR^2 , the improvement over a heavy-atom-count null.

Keywords: topological index; QSPR; molecular descriptor; degree-based index; null baseline; chemical graph theory

1 Introduction

A molecular graph G represents a hydrogen-suppressed molecule with atoms as vertices and bonds as edges. A *degree-based topological index* assigns a real number to G by summing a symmetric function of the endpoint degrees over all edges,

$$\text{TI}_f(G) = \sum_{uv \in E(G)} f(d_u, d_v).$$

Instances include the Zagreb indices (Gutman & Trinajstić, 1972), the Randić connectivity index (Randić, 1975), the atom–bond connectivity (ABC) and geometric–arithmetic (GA) indices, the forgotten index (Furtula & Gutman, 2015), and the Sombor index (Gutman, 2021). Several thousand such invariants have been proposed.

1.1 The template under examination

A recognisable publication template has become common in pharmacy, pharmaceutical-chemistry and applied-mathematics journals: select a therapeutic class and 10–25 drug molecules; compute 6–20 degree-based indices; obtain 5–13 physicochemical properties, typically from an online compilation; regress each index against each property; and report, per property, the index with the largest R^2 , described as the best predictor. The properties most often used are molecular weight, molar refractivity, molar volume, polarizability, surface tension, boiling point and flash point.

1.2 The difficulty

Four of those properties — molecular weight, molar refractivity, molar volume, polarizability — are *additive*: each is, to good approximation, a sum of atomic or group contributions, and hence a near-linear function of molecular size. Degree-based indices are also near-linear functions of size. A high R^2 between the two is therefore close to guaranteed and carries little information about the index.

This is not a new suspicion. Gutman & Tošović (2013) tested 20 vertex-degree-based indices against heats of formation and normal boiling points of *octane isomers* and found the correlation ability of many to be weak or nil, with augmented Zagreb and ABC performing best. Their design fixes n and m , removing size variation entirely — and, we note, forcing the choice of non-additive properties, since molecular weight and molar refractivity are literally constant within a constitutional-isomer class. That paper is cited in a large fraction of the studies described above, often as introductory boilerplate, without its implication being drawn.

Randić et al. (1994) had earlier addressed descriptor intercorrelation directly, arguing that apparent parallelism need not be a concern unless descriptors are strictly collinear, and that orthogonalization reveals genuine structural differences even between nearly collinear descriptors. We accept this as stated. We observe only that it concerns *orthogonalized combinations*. It provides no licence for the comparative claim — “index X is the best predictor of property Y ” — which the template produces.

2 Theory

2.1 Profile decomposition

Let $P = \{(i, j) : 1 \leq i \leq j \leq \Delta\}$ be the set of unordered degree pairs available at maximum degree Δ (for hydrogen-suppressed organic molecules $\Delta = 4$, so $|P| = 10$). For a molecular graph G with m edges define the *edge-degree-pair profile* $p(G) \in \Delta(P)$ by

$$p_\pi(G) = \frac{\#\{uv \in E(G) : \{d_u, d_v\} = \pi\}}{m},$$

a point in the probability simplex over P . Regarding f as a vector in $\mathbb{R}^P$, every degree-based index factorises exactly:

$$\text{TI}_f(G) = m(G) \cdot \langle p(G), f \rangle. \quad (1)$$

The index is the product of an *extensive* factor (edge count) and an *intensive* factor (mean edge weight). All structural information beyond size is carried by $\langle p(G), f \rangle$.

2.2 The bound

Consider molecules $G_1, \dots, G_N$ with edge counts m_i and profiles p_i . Let $\bar{p} = N^{-1} \sum_i p_i$, $u_i = p_i - \bar{p}$, $\bar{g} = \langle \bar{p}, f \rangle > 0$ and $\text{CV} = \text{sd}(m)/\text{mean}(m)$. Write $\tilde{v}$ for the mean-centred version of $v \in \mathbb{R}^N$.

Proposition 1. Let $E_i = m_i \langle u_i, f \rangle$ and $\tau_f = \|\tilde{E}\|/(\bar{g}\|\tilde{m}\|)$. If $\tau_f \leq 1$ then

$$r^2(\text{TI}_f, m) \geq 1 - \tau_f^2. \quad (2)$$

Moreover, with $\Sigma = \sum_i w_i u_i u_i^\top$, $w_i = m_i^2 / \sum_j m_j^2$, and $f_c = f - c\mathbf{1}$ for any scalar c ,

$$\tau_f \leq \frac{\sqrt{f_c^\top \Sigma f_c} \text{rms}(m)}{\bar{g} \text{sd}(m)}. \quad (3)$$

Proof. From (1), $\text{TI}_f(G_i) = \bar{g}m_i + E_i$. Set $S = \bar{g}m$, so $\text{TI}_f = S + E$ and, centring being linear, $\widetilde{\text{TI}_f} = \tilde{S} + \tilde{E}$ with $\tilde{S} = \bar{g}\tilde{m}$ parallel to $\tilde{m}$.

For (2), note $r(\text{TI}_f, m) = \cos \angle(\widetilde{\text{TI}_f}, \tilde{m}) = \cos \angle(\widetilde{\text{TI}_f}, \tilde{S})$. The set $\{\tilde{S} + e : \|e\| \leq \rho\}$ is a Euclidean ball of radius ρ centred at $\tilde{S}$. Every point of that ball lies within angle $\arcsin(\rho/\|\tilde{S}\|)$ of $\tilde{S}$ when $\rho < \|\tilde{S}\|$, the extreme being attained on the tangent line from the origin. Taking $\rho = \|\tilde{E}\|$ gives $\angle(\widetilde{\text{TI}_f}, \tilde{S}) \leq \arcsin \tau_f$ and hence $r \geq \sqrt{1 - \tau_f^2}$.

For (3): each u_i satisfies $\langle u_i, \mathbf{1} \rangle = 0$, so $\langle u_i, f \rangle = \langle u_i, f_c \rangle$ for every c — the weighting function is determined only up to an additive constant. Centring is an orthogonal projection, so $\|\tilde{E}\| \leq \|E\|$, and

$$\|E\|^2 = \sum_i m_i^2 \langle u_i, f_c \rangle^2 = \left(\sum_j m_j^2 \right) f_c^\top \Sigma f_c,$$

whence $\|E\| = \sqrt{N} \text{rms}(m) \sqrt{f_c^\top \Sigma f_c}$. Dividing by $\bar{g}\|\tilde{m}\| = \bar{g}\sqrt{N} \text{sd}(m)$ gives (3). $\square$

2.3 A corollary over all properties

The Proposition constrains the relationship between TI_f and edge count; the claim of interest concerns properties. The gap closes in one step.

Lemma 1. For any random variables X, Y, Z , $|r(X, Z) - r(Y, Z)| \leq \sqrt{2(1 - r(X, Y))}$.

Proof. Let x, y, z be the centred, unit-normalised data vectors. Then $r(X, Z) - r(Y, Z) = \langle x - y, z \rangle \leq \|x - y\| \|z\| = \|x - y\|$, and $\|x - y\|^2 = 2 - 2\langle x, y \rangle = 2(1 - r(X, Y))$. Symmetry gives the absolute value. $\square$

Corollary 1. Let $\rho = r(\text{TI}_f, m) \geq \sqrt{1 - \tau_f^2}$ by (2). Then for every property P whatsoever,

$$|r(\text{TI}_f, P) - r(m, P)| \leq \sqrt{2(1 - \rho)} \leq \sqrt{2(1 - \sqrt{1 - \tau_f^2})}. \quad (4)$$

This is the operative statement. It is not a claim about the properties we happened to test: for any response variable at all — measured, computed, physicochemical, biological — a degree-based index and a bond count are interchangeable to within a margin fixed by τ_f , a property of the index and the molecule set alone. For ABC on our melting-point set, $\rho = 0.9983$ and the margin is 0.058; across 14 indices and 5 properties there are no violations and the largest observed deviation for ABC is 0.015.

2.4 Interpretation

The bound factorises into three parts, each with a chemical reading.

- $\sqrt{f_c^\top \Sigma f_c}$ — **index spread along realised variation:** not how much f varies over all conceivable degree pairs, but how much it varies along the directions in which real molecular profiles actually differ. This is why the naive sandwich $f_{\min}m \leq \text{TI}_f \leq f_{\max}m$, which uses the worst-case range of f , is far too weak: it ignores the covariance structure entirely.

- $1/\bar{g}$ — **normalisation**. Only the relative spread of f matters.
- $\text{rms}(m)/\text{sd}(m) \approx \sqrt{1 + \text{CV}^2}/\text{CV}$ — **size dispersion**, which diverges as $\text{CV} \rightarrow 0$.

These are the complete list of things one could break to construct a degree-based index that is not a size proxy. Two are properties of the index; the third is a property of the *study design*, and is under the analyst’s control.

The CV term recovers Gutman & Tošović (2013) immediately. On an isomer set $\text{CV} = 0$ exactly, the bound is vacuous, and the indices are free to differ arbitrarily from edge count and from one another — precisely the regime in which comparing them is meaningful. Their study and the studies of §1.1 were never in conflict; they sit at opposite ends of a continuum that (2)–(3) parameterises.

3 Methods

Index battery. Fourteen degree-based indices with weighting functions $f(a, b)$: first Zagreb $a + b$; second Zagreb ab ; forgotten $a^2 + b^2$; hyper-Zagreb $(a + b)^2$; Randić $(ab)^{-1/2}$; sum-connectivity $(a + b)^{-1/2}$; ABC $\sqrt{(a + b - 2)/(ab)}$; geometric–arithmetic $2\sqrt{ab}/(a + b)$; harmonic $2/(a + b)$; Sombor $\sqrt{a^2 + b^2}$; augmented Zagreb $(ab/(a + b - 2))^3$; inverse sum indeg $ab/(a + b)$; symmetric division $\deg a/b + b/a$; redefined third Zagreb $ab(a + b)$.

Datasets. A hand-curated drug set ($N = 38$) from the therapeutic classes recurring in the target literature; a random subsample of a drug-like ZINC collection ($N = 6,000$); ESOL measured aqueous solubility ($N = 1,105$, Delaney, 2004); Open Notebook Science melting points ($N = 8,583$); and a boiling-point compilation ($N = 6,400$). Structures for the last two were joined by CAS lookup through PubChem. Filters: 10–60 heavy atoms, maximum degree ≤ 4 , must contain carbon, salts and mixtures excluded.

Null baseline. The null descriptor is the **heavy-atom count** n . For each property we report $R^2(n)$, R^2 of the best of the 14 indices, and $\Delta R^2 = R^2(\text{best}) - R^2(n)$. Out-of-sample performance uses repeated 70/30 splits. Small-sample behaviour uses repeated random draws of 15 molecules, matching the modal published study size.

4 Results

4.1 The indices span one dimension

Table 1: Dimensionality of the 14-index battery across two chemical spaces.

	drug set ($N = 38$)	ZINC ($N = 6,000$)
PC1 variance explained	98.64%	97.49%
PC2	1.13%	1.76%
PC3	0.21%	0.69%
median pairwise $ r $	0.989	0.979
minimum pairwise $ r $	0.934	0.908
min $r(\text{TI}, m)$	0.961	0.947

The result replicates across unrelated chemical spaces and is not an artefact of size range: the ZINC sample has roughly half the size dispersion of the drug set ($\text{CV}(m) = 0.213$ vs 0.408) yet shows the same collapse. A rank statement makes the point exactly. On ESOL, cross-validated R^2 for measured $\log S$: n alone 0.3199; best single index 0.3348; all 14 indices 0.3803; n plus

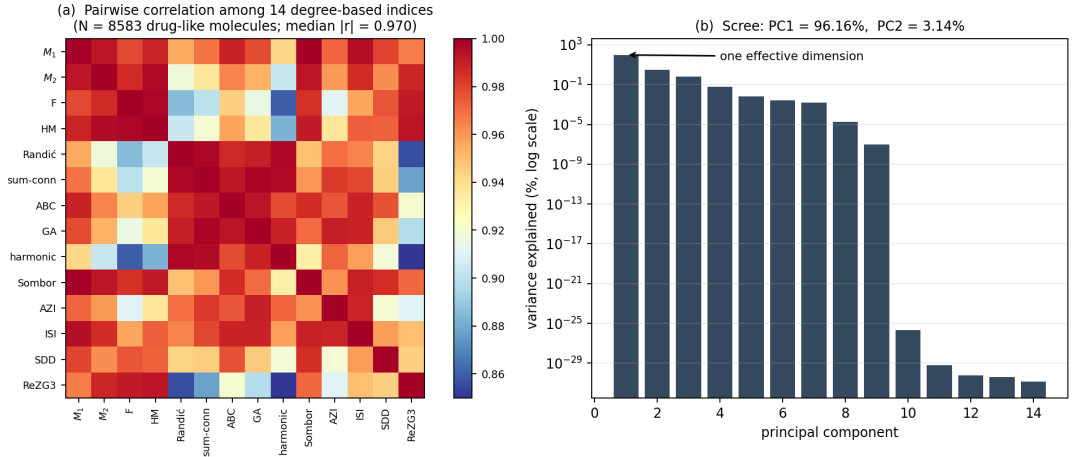


Figure 1: (a) Pairwise correlations among the 14 indices on 8,583 drug-like molecules. (b) Scree plot on a logarithmic axis: the battery has one effective dimension.

all 14 indices 0.3803 — identical to four decimal places. Heavy-atom count already lies in their span.

The observed edge-degree-pair profile is highly concentrated: across the drug set four of the ten available pairs account for 91% of all edges, namely (2, 3) 41.3%, (2, 2) 17.7%, (3, 3) 17.5% and (1, 3) 14.2%.

4.2 The bound is valid and tight

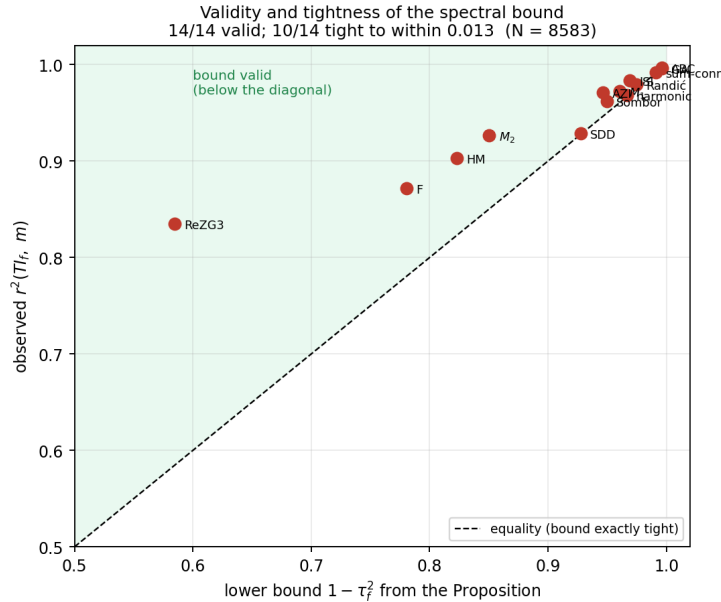


Figure 2: Spectral bound (3) against observed $r^2(TI_f, m)$ on the melting-point set. All 14 points lie above the diagonal (bound valid); ten lie within 0.013 of it. Per-index values are given in the Appendix.

The spectral bound is valid in 14/14 cases and non-vacuous in 14/14. A Hölder bound using $\|u_i\|_1$ and the range of f is valid but vacuous ($\tau > 1$) for 6 of the 14 indices: the covariance structure is essential.

4.3 Against additive properties, no index beats counting atoms

We anchor on **molecular weight**, which is exactly additive and is not the output of any fitted model. Molar refractivity (Crippen) and Labute ASA are themselves fitted additive schemes, so correlations with them are partly circular; we report them for comparability with the literature but they carry no independent weight.

Table 2: Null-baseline comparison on additive properties. Negative ΔR^2 means the heavy-atom count beats every one of the 14 indices.

dataset	property	$R^2(n)$	$R^2(\text{best})$	ΔR^2
Drug set (38)	molecular weight	0.9850	0.9826	-0.0025
	Labute ASA	0.9911	0.9930	+0.0019
	molar refractivity	0.9564	0.9696	+0.0131
ESOL (1,105)	molecular weight	0.8510	0.8645	+0.0136
	molar refractivity	0.9414	0.9432	+0.0018
	Labute ASA	0.9618	0.9521	-0.0096
T_m set (8,583)	molecular weight	0.7977	0.7941	-0.0036
	molar refractivity	0.8856	0.9128	+0.0272
T_b set (6,400)	molecular weight	0.8430	0.8136	-0.0295
	molar refractivity	0.8233	0.8824	+0.0591

In four of ten cases the null beats every index outright; the largest gain anywhere is +0.059.

4.4 Against measured non-additive properties

No single index exceeds $R^2 = 0.212$ on a measured non-additive property. We state the countervailing result plainly: used *jointly*, the 14 indices do better than any of them alone.

Table 3: Cross-validated R^2 for measured melting point ($N = 8,583$).

model	CV R^2
heavy-atom count n	0.0713
molecular weight	0.0626
best single index (ReZG3)	0.1545
n + ring count	0.2179
all 14 indices	0.3364
all 14 indices + n + ring count	0.3359

Two things follow. The best single index is beaten by a two-number baseline requiring no chemical graph theory at all. And the 14 indices jointly *do* carry information beyond simple counts (0.336 vs 0.218) — a real effect we do not dispute — but adding n and ring count to the battery changes cross-validated performance by -0.0005 , confirming that trivial size descriptors already lie within their span. The joint model is a legitimate descriptor set; the *comparative single-index ranking* that the template produces is not a legitimate inference.

4.5 Reported R^2 is a readout of sample size dispersion

Nothing changes across a row of Figure 3 except how much size variation the sample contains. As the headline R^2 for molar refractivity climbs from 0.34 to 0.94, the index’s contribution over atom count collapses from +0.148 to +0.005: *the most impressive-looking results are those in which the descriptor is doing the least work.*

At published sample sizes the winner is unstable — all 14 indices win at some frequency, the modal winner taking 42–48% for melting and boiling point and as little as 14% at low size

Reported R^2 in degree-based-index QSPR is a function of how much size variation the sample contains (ESOL, subsamples of 15 molecules — the typical published study size)

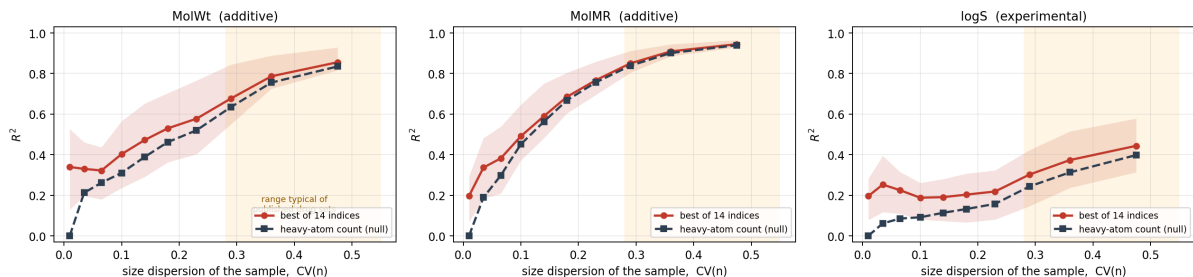


Figure 3: Reported R^2 as a function of the size dispersion of the sample, for two additive properties and one measured property. Subsamples of 15 molecules from ESOL. Red: best of 14 indices; black dashed: heavy-atom count null. Shaded band: interquartile range. Orange region marks the dispersion typical of published drug sets.

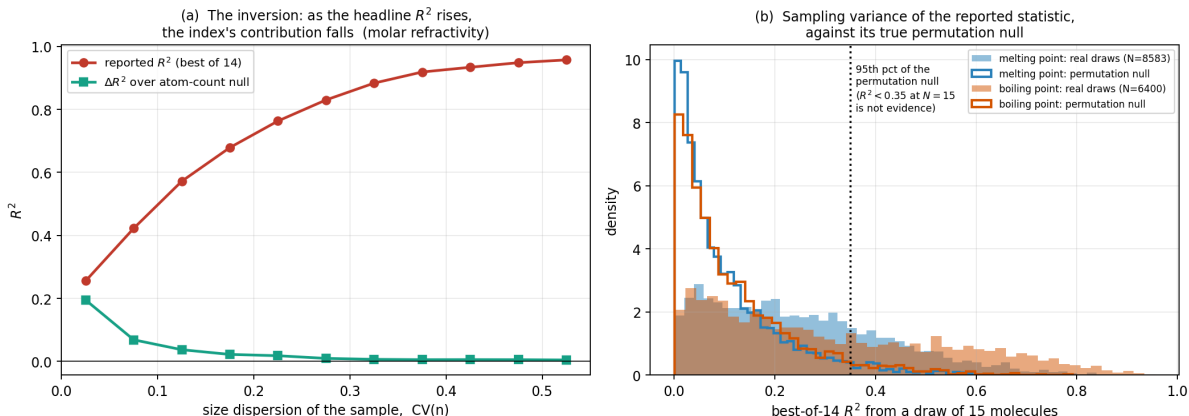


Figure 4: (a) The inversion: as the headline R^2 rises with size dispersion, the improvement over the atom-count null falls toward zero. (b) Sampling distribution of the reported statistic (filled) against its true permutation null (outline).

dispersion. Contrasting the sampling distribution with the permutation null (Figure 4b) gives two conclusions we are careful to separate. **(i)** Any best-of-14 R^2 below roughly 0.35 at $N = 15$ falls inside the permutation null and is not evidence of a structure–property relationship; a substantial share of published values lies in this range. **(ii)** Values well above that are *not* explicable as noise — they are explicable by upward bias. At $N = 15$ the expected best-of-14 R^2 is 0.306 against a full-sample value of 0.212, with a 5th–95th range of 0.026–0.730. A single reported value from a 15-molecule study is compatible with almost any underlying effect size, and the publication filter selects the upper tail. We do not claim published results are noise; we claim they are uninformative as point estimates.

4.6 Functional form: the threshold is not fixed

Studies in the target literature frequently fit quadratic or cubic rather than linear models. We repeated the analysis at polynomial degrees 1–3, fitting the *same* order to the index and to the null.

At full sample size the verdict is unchanged, with one honest exception. For molecular weight ΔR^2 is -0.0036 , -0.0036 and -0.0041 at degrees 1, 2 and 3: the null beats every index at every order. For boiling point ΔR^2 is likewise flat ($+0.060$, $+0.057$, $+0.057$). For melting

point, however, the indices benefit from curvature more than atom count does and ΔR^2 rises from +0.084 to +0.112; our linear-only analysis was mildly conservative there. In absolute terms both remain poor ($R^2 = 0.085$ for the null against 0.197 for the best index at degree 3), and cross-validated values track in-sample values to within 0.001 at $N = 8,583$.

At $N = 15$ the effect is dramatic. Fitting a cubic means estimating four parameters from fifteen points, and the permutation null inflates accordingly.

Table 4: Permutation null and real-draw distributions of best-of-14 R^2 at $N = 15$.

property	fit	null mean	null 95th	real mean	real 95th
melting point	linear	0.104	0.323	0.248	0.582
	quadratic	0.207	0.475	0.351	0.664
	cubic	0.316	0.605	0.442	0.729
boiling point	linear	0.117	0.358	0.311	0.734
	quadratic	0.237	0.533	0.435	0.782
	cubic	0.349	0.654	0.543	0.848

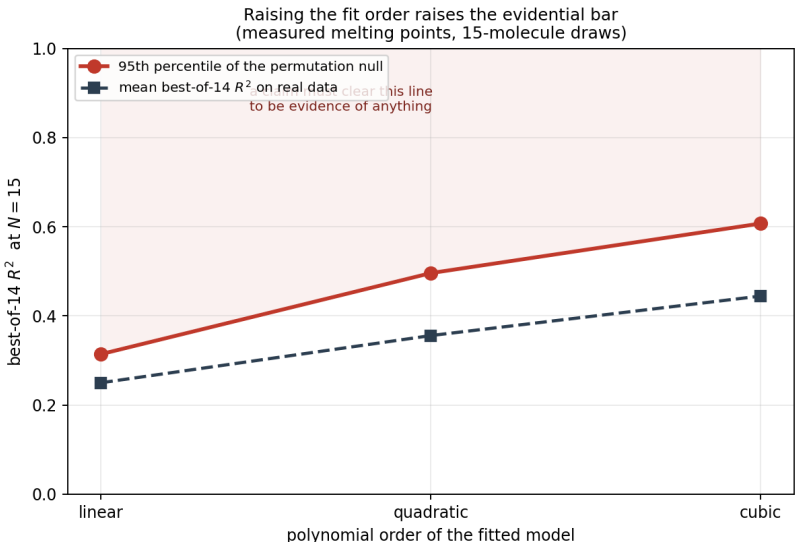


Figure 5: Raising the polynomial order of the fitted model raises the evidential bar. A claim must clear the red line to be distinguishable from a permutation null at the same sample size and fit order.

The evidential threshold is therefore *not a fixed number but a function of the fit order the study used*. A paper reporting $R^2 = 0.60$ from a cubic fit to 15 molecules has produced a value lying inside its own permutation null; the same value from a linear fit would be well outside it. Higher-order fits do not strengthen a small-sample QSPR claim — they raise the bar it must clear, and papers reporting improved R^2 after moving from linear to cubic at $N \approx 15$ are reporting overfitting.

5 Discussion

5.1 What is and is not claimed

We do **not** claim degree-based indices carry no information. A selection-bias null (shuffled response, maximum over 14 indices) gives mean $R^2 = 0.0011$ and 95th percentile 0.0040 on

ESOL, against an observed ΔR^2 of +0.0143: a small but real margin. Out of sample the indices genuinely help on some additive properties.

What we claim is narrower: **the comparative claims produced by this template are not interpretable**, because the objects being compared span one dimension, the properties most often used are size proxies, and the sample sizes used cannot distinguish among 14 collinear predictors.

5.2 Reconciliation with prior work

Gutman & Tošović (2013) are not contradicted; they are explained. Their fixed- n design sets $CV = 0$, makes the bound vacuous, and creates the only conditions under which the indices can be told apart — which is why they found real differences. The subsequent literature restored size variation, adopted additive properties, and reported the resulting size correlation as predictive success. Randić et al. (1994)’s orthogonalization argument stands and licenses continued interest in intercorrelated descriptors *used in combination*; it does not license single-index comparative rankings. Roy & Gramatica (2014)’s warning about R^2 concerns external validation and applies here with additional force, but is orthogonal to the size-proxy mechanism.

5.3 A checklist for editors and referees

The following can be applied without specialist knowledge of chemical graph theory, and instantiates rather than extends existing QSAR validation practice.

1. **Report ΔR^2 over a heavy-atom-count null** for every index–property pair. If $\Delta R^2 \approx 0$, the result concerns molecular size and should be described that way.
2. **Report $CV(n)$** for the molecule set. Without it, R^2 is uninterpretable.
3. **State whether the property is additive.** Molecular weight, molar refractivity, molar volume and polarizability are; correlations with them are not evidence about an index.
4. **Do not crown a best index at $N < 30$** without a resampling analysis of winner stability.
5. **Compare R^2 against a permutation null computed at the study’s own sample size and fit order.** Indicative 95th percentiles at $N \approx 15$: linear ≈ 0.35 , quadratic ≈ 0.50 , cubic ≈ 0.63 . Raising the fit order raises this bar.
6. **Report the provenance of property values** — measured, or estimated by an additive group-contribution scheme. Regressing an additive estimate on an additive descriptor is circular.

5.4 Limitations

The boiling-point compilation is a handbook aggregation that may include estimated as well as measured values; the melting-point set is curated experimental data and should be treated as primary. The CAS-to-structure join through PubChem has an unmeasured mismatch rate. The maximum-degree ≤ 4 filter, which we had expected to bias the sample, excludes exactly one molecule of 8,584. Our battery covers degree-based indices only — distance-based, eccentricity-based and neighbourhood-degree-based families are not tested, though (1) suggests analogous arguments. The drug set of 38 molecules was assembled by hand and is not a random sample of drug space; all headline claims are replicated on the larger unbiased sets.

Declaration of generative AI use

Portions of this work were carried out with the assistance of Claude, a large language model developed by Anthropic. Its contribution was substantial and is described here in full so that

readers and referees can weight it appropriately.

The model was used to: survey the prior literature and identify the relevant precedents, including Gutman & Tošović (2013) and Randić et al. (1994); propose and formalise the profile decomposition (1) and the bound (2)–(3); identify and prove the corollary (4); write all analysis and figure-generation code; execute the computational experiments reported in Section 4; and draft substantial portions of the manuscript text.

The research direction, the choice of which lines of enquiry to pursue and abandon, and all decisions about scope, framing and claim strength were made by the author. Two of the model’s initial hypotheses — a worst-case sandwich bound, and a claim that heavy-atom count would outperform every index out of sample — were falsified during the work and retracted; the results reported here reflect the surviving claims. An adversarial review pass, also model-assisted, identified and corrected a mischaracterised null distribution and one factual error in an earlier draft.

The author has verified the reported results, takes full responsibility for the content of this paper, and affirms that the generative AI tool is not and cannot be an author. All code and data required to reproduce every number and figure are provided as supplementary material, and independent reproduction is encouraged.

Reproducibility

All datasets are public. Analysis code, molecule lists and figure scripts are provided as supplementary material; the full pipeline runs in a few minutes on a laptop using RDKit and NumPy.

A Per-index bound values

Table 5: Spectral bound against observed $r^2(\text{TI}_f, m)$, melting-point set ($N = 8,583$, $\text{CV}(m) = 0.417$). Sorted by slack.

index	τ (exact)	τ (bound)	$1 - \tau_{\text{spec}}^2$	observed r^2
ABC	0.0605	0.0608	0.9963	0.99663
GA	0.0613	0.0614	0.9962	0.99648
sum-connectivity	0.0924	0.0929	0.9914	0.99231
SDD	0.2680	0.2686	0.9279	0.92885
harmonic	0.1817	0.1828	0.9666	0.96868
Randić	0.1567	0.1585	0.9749	0.97974
M_1	0.1975	0.1986	0.9606	0.97271
Sombor	0.2235	0.2242	0.9497	0.96221
ISI	0.1739	0.1761	0.9690	0.98370
AZI	0.2289	0.2316	0.9464	0.97101
M_2	0.3836	0.3867	0.8504	0.92675
HM	0.4183	0.4204	0.8232	0.90313
F	0.4669	0.4683	0.7807	0.87174
ReZG3	0.6405	0.6447	0.5844	0.83514

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